TOPOLOGICAL INSULATORS — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: March 29, 2026

Diagram-based questions on Topological Insulators

Question 1

10.0 points

Analyze the band structure diagram of a topological insulator shown in the diagram below, which features band inversion between the bulk valence and conduction bands. Explain how this inversion leads to a non-trivial topology.

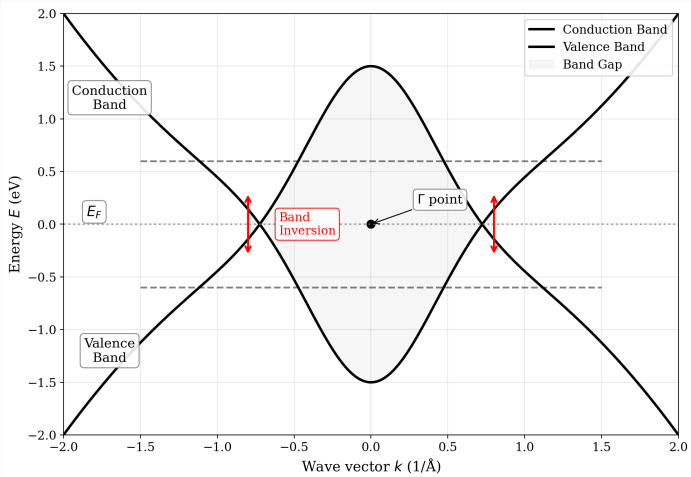

Fig. 1. Diagram for Question 1

Question 2

10.0 points

Interpret the bulk-boundary correspondence as depicted in the surface state diagram of a topological insulator shown in the diagram below. Describe how the surface states are protected by topology.

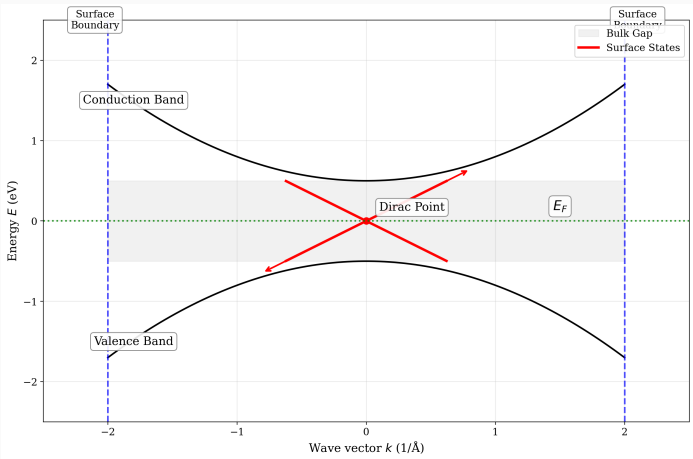

Fig. 2. Diagram for Question 2

Question 3

10.0 points

Analyze the edge state diagram below associated with the quantum spin Hall effect. Explain how these states allow for dissipationless edge transport.

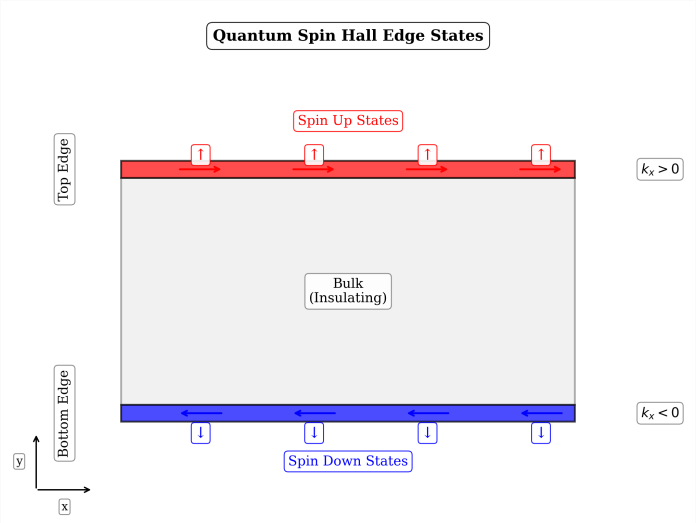

Fig. 3. Diagram for Question 3

Question 4

10.0 points

Analyze the k-space structure of a 2D system with a Z2 topological invariant as depicted in the diagram below. Explain the significance of the Z2 invariant.

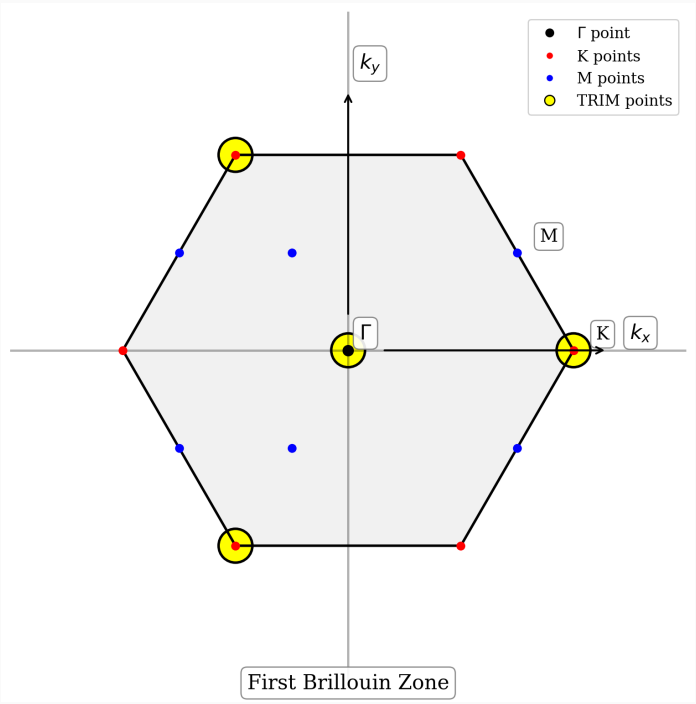

Fig. 4. Diagram for Question 4

Question 5 10.0 points

Examine the Dirac cone dispersion relation depicted in the diagram below on the surface of a 3D topological insulator. Discuss how this structure is related to massless Dirac fermions.

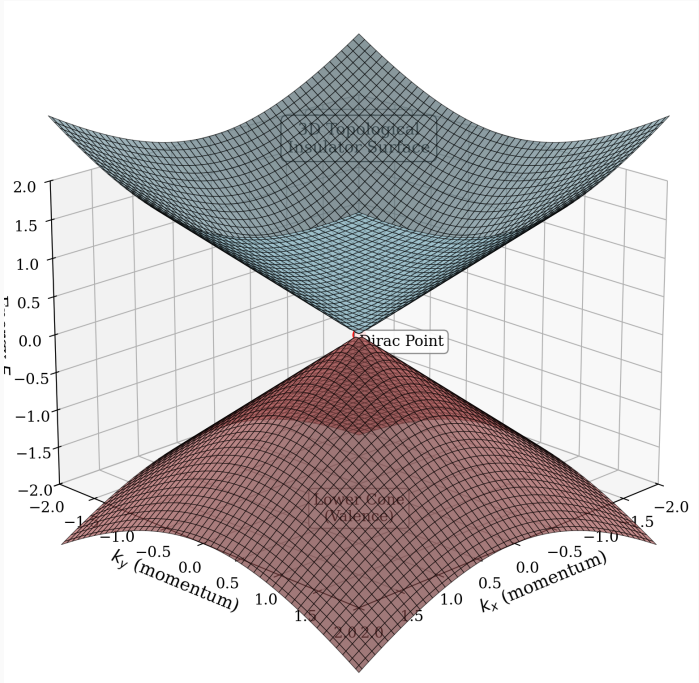

Fig. 5. Diagram for Question 5